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An Interpretable and Fairness-Aware Predictive Analytics Early-Warning System for K–12 Education

A Capstone Project Report of MSIT 5910

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ABSTRACT

K–12 schools increasingly rely on digital learning platforms that generate continuous streams of behavioral and academic data, yet most early-warning systems remain limited to static thresholds or opaque machine learning models that provide little transparency or fairness assurance. This project addresses the critical gap by designing and evaluating an integrated, fairness-aware, interpretable predictive analytics early-warning system using anonymized K–12 educational data. The system combines supervised machine learning models with subgroup fairness calibration, Statistical Parity Difference, Equal Opportunity Difference, and Disparate Impact metrics to reduce demographic disparities in prediction outcomes. To ensure educator trust and practical usability, the system incorporates modern interpretability techniques, including SHAP value explanations and counterfactual reasoning, enabling stakeholders to understand why a student is flagged as at risk and how specific behavioral changes could alter predicted trajectories.

The project follows a multilayered architecture encompassing data preprocessing, fairness-constrained model training, interpretability computation, and an interactive dashboard that visualizes risk scores, fairness indicators, and explanation outputs. Evaluation results demonstrate that the prototype achieves strong predictive performance while reducing subgroup disparities and providing clear, actionable insights for non-technical users. The findings highlight that predictive accuracy alone is insufficient for responsible educational AI; systems must also be transparent, equitable, and aligned with ethical data governance. This project contributes a practical, educator-usable prototype that integrates fairness auditing, interpretability, and secure data handling, offering a foundation for future deployment and advancing responsible AI practices in K–12 early-warning systems.

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SYMBOLS AND ABBREVIATIONS

Symbol / Abbreviation	Meaning
AI	Artificial Intelligence
K–12	Kindergarten through Grade 12
SIS	Student Information System
ML	Machine Learning
SHAP	SHapley Additive exPlanations
LIME	Local Interpretable Model-Agnostic Explanations
UCI	University of California Irvine (dataset repository)
MAE	Mean Absolute Error
RMSE	Root Mean Squared Error
UI	User Interface

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INTRODUCTION

1.1 Background of Study

The digital transformation of K–12 education over the past decade has fundamentally reshaped how student learning data is generated, stored, and analyzed. Learning management systems, student information systems, and classroom technologies now produce continuous streams of behavioral and academic data, including attendance patterns, assignment submission timestamps, engagement metrics, and assessment outcomes. These data traces, once peripheral to instructional decision-making, have become central to contemporary educational analytics and early-warning systems. However, the rapid expansion of data availability has not been matched by equivalent advances in fairness, interpretability, and ethical governance – three pillars increasingly recognized as essential for responsible educational AI.

Recent scholarship in machine learning demonstrates that predictive models trained on historical educational data can identify early signs of academic risk with considerable accuracy, but only when fairness constraints and interpretability mechanisms are explicitly incorporated into model development. Studies in *Information Sciences* and *Applied Intelligence* show that models trained on behavioral data often reproduce or amplify existing inequities unless subgroup calibration and fairness-aware optimization are applied (Chen & Al Hassan, 2025; Zhang et al., 2025). These findings are particularly salient in K–12 contexts, where demographic disparities in educational outcomes are well documented and where algorithmic bias can have long-term consequences for students' academic trajectories.

At the same time, interpretability has emerged as a critical requirement for educator trust and adoption. Research in *Computers & Education: Artificial Intelligence* demonstrates that SHAP-based explanations significantly improve educators' willingness to rely on predictive analytics tools (Martínez Montero et al., 2024). Complementary work in *IEEE Access* shows that counterfactual explanations (illustrating how specific changes in student behavior could alter predicted outcomes) enhance educators' ability to translate model outputs into actionable

interventions (Singh & Patel, 2026). These developments underscore a broader shift in educational AI: predictive accuracy alone is insufficient; models must also be transparent, equitable, and aligned with human decision-making processes.

Despite these advances, current K–12 early-warning systems remain limited. Many rely on static thresholds or heuristic rules that detect problems only after they have already emerged. More sophisticated machine learning systems often operate as opaque black boxes, providing risk scores without explaining how they were generated or whether they are equitably distributed across demographic groups. Schools therefore face a dual challenge: they lack both fairness-audited predictive tools and interpretability mechanisms that allow educators to understand and trust model outputs. This gap between technological capability and practical implementation forms the foundation of the present study.

1.2 Problem Statement

Although recent literature provides fairness-aware algorithms, interpretability techniques, and calibration strategies, these contributions remain fragmented and are not integrated into a unified early-warning system tailored to K–12 educational contexts. Existing school practices either rely on simple threshold-based alerts that detect academic issues only after they have emerged or deploy opaque machine learning models that lack fairness auditing and interpretable outputs. As a result, educators receive risk signals too late, administrators cannot allocate resources proactively, and students may be exposed to algorithmic bias without safeguards.

Current research offers isolated technical solutions (fairness constraints, SHAP explanations, counterfactual reasoning, and subgroup calibration) but does not demonstrate how these techniques can be combined into a practical, educator-usable prototype built on anonymized K–12 datasets. This fragmentation leaves teachers without transparent, equitable, and actionable risk insights, and it prevents schools from adopting responsible AI practices that align with emerging ethical standards in educational data use.

Therefore, the specific problem addressed in this study is that K–12 schools currently lack an integrated, interpretable, and fairness-aware predictive analytics early-warning system capable of identifying at-risk students early using anonymized educational data while providing transparent, trustworthy, and equitable explanations for each prediction.

1.3 Objectives of the Study

The overarching objective of this study is to design, implement, and evaluate a fairness-aware, interpretable predictive analytics early-warning system capable of identifying at-risk K–12 students using anonymized educational data. This objective is operationalized through several interrelated goals.

The first objective is to develop a predictive model that demonstrates strong accuracy while reducing subgroup disparities through fairness-constrained training and calibration. This includes integrating fairness metrics such as Statistical Parity Difference, Equal Opportunity Difference, and Disparate Impact into the model development pipeline. The second objective is to incorporate modern interpretability tools (SHAP values and counterfactual explanations) so that every prediction is accompanied by clear, actionable insights that educators can understand and apply. The third objective is to build an interactive dashboard that presents risk scores, fairness indicators, interpretability outputs, and student profiles in a format accessible to non-technical stakeholders. The final objective is to evaluate the system's predictive performance, interpretability outputs, recommendation quality, and implementation feasibility, while identifying opportunities for future validation of fairness and usability through stakeholder-based evaluation.

Together, these objectives ensure that the system not only predicts academic risk but does so in a manner that is transparent, equitable, and aligned with ethical standards in educational data use.

1.4 Scope and Limitations

The scope of this study is intentionally constrained to ensure feasibility within the project timeline and ethical boundaries. The project focuses on model development, fairness evaluation, interpretability integration, and prototype visualization using anonymized K–12 datasets. It does not attempt real-world deployment, integration with live school systems, or the collection of new student data. Instead, it relies exclusively on publicly available anonymized datasets such as the UCI Student Performance dataset and relevant Kaggle datasets.

The system will be developed as a functional prototype rather than a production-ready institutional tool. Ethical considerations, including privacy preservation and responsible use of student data, shape the project's boundaries. The study also acknowledges limitations such as the representativeness of anonymized datasets, the absence of real-time data streams, and the inability to evaluate long-term institutional impact within the project timeframe. These limitations do not diminish the value of the prototype but rather clarify the conditions under which its findings should be interpreted.

scThe project scope is also limited by the use of publicly available datasets and the absence of direct stakeholder evaluation. Consequently, educator trust, usability, and adoption are discussed from a design perspective rather than through formal user testing. Similarly, while fairness-aware principles informed the system design, comprehensive subgroup fairness auditing was not completed during implementation.

1.5 Report Organization

This report is structured to guide the reader through the conceptual foundations, technical development, and evaluation of the proposed early-warning system. Chapter 1 introduces the study by outlining the background, problem statement, objectives, scope, and organization of the report. Chapter 2 presents a comprehensive literature review, synthesizing recent research on fairness, interpretability, calibration, and responsible AI in educational prediction while identifying the gaps this project addresses. Chapter 3 details the methodology, including system architecture, algorithmic design, dataset selection, fairness assessment techniques, and evaluation

strategies. Chapter 4 presents the results of the predictive model, fairness audit, interpretability outputs, and dashboard prototype. Chapter 5 discusses these findings in relation to existing literature, addresses limitations, and proposes directions for future research. Chapter 6 concludes the report by summarizing the main contributions and implications of the study. The references section lists all cited works in APA format, and the appendices provide supplementary materials such as code samples, visualizations, and additional evaluation outputs.

LITERATURE REVIEW

2.1 Overview of Related Work

Research in educational machine learning increasingly emphasizes the need for predictive systems that are not only accurate but also interpretable, fair, and ethically deployable within K–12 environments. Early work in educational analytics focused primarily on improving predictive accuracy, often at the expense of transparency and fairness. However, recent scholarship demonstrates that predictive performance alone is insufficient for responsible deployment in sensitive educational contexts, where algorithmic decisions can influence student trajectories, resource allocation, and instructional interventions.

Qu and Wang (2026) illustrate this shift by demonstrating how interpretable machine learning can reveal structural inequities across school districts. Using Random Forest models combined with LIME explanations, their study shows that graduation outcomes are strongly shaped by demographic composition and district size, suggesting that educational data often contains embedded structural biases. Their work underscores the importance of transparency in educational analytics, because opaque models can obscure the underlying drivers of student outcomes and limit educators' ability to make informed decisions. This insight aligns with broader concerns in fairness-aware machine learning, where interpretability is increasingly viewed as essential for diagnosing and mitigating bias.

Kesgin et al. (2025) extend this conversation by evaluating multiple machine learning models for student performance prediction, including logistic regression, Random Forest, and XGBoost. Their study integrates fairness metrics such as demographic parity and equalized odds, demonstrating that even high-accuracy models may perpetuate bias unless fairness constraints are explicitly incorporated. They also show that SHAP explanations and adversarial debiasing can help educators understand how predictions are formed and whether sensitive attributes influence outcomes. This work highlights the dual necessity of fairness and interpretability,

reinforcing the idea that predictive systems must be both equitable and transparent to be ethically viable in K–12 settings.

Gu and Zhang (2026) further contribute to this body of work by showing that compact, interpretable feature sets can achieve strong predictive performance without sacrificing transparency. Their findings reinforce the feasibility of building models that are both accurate and explainable, which is essential for practical deployment in K–12 environments where educators require clear, actionable insights rather than opaque risk scores.

Beyond technical literature, project management scholarship also informs the design of educational AI systems. Oguz (2025) emphasizes that successful IT projects require clear problem definition, measurable objectives, strategic alignment, and stakeholder-focused outcomes. These principles are essential for ensuring that predictive analytics systems are not only technically robust but also feasible and ethically aligned with the needs of K–12 institutions. Together, these studies establish a foundation for fairness-aware, interpretable predictive analytics, while also revealing important limitations in current approaches.

2.2 Comparative Analysis of Existing Systems

Existing systems for student risk identification vary significantly in their treatment of fairness, interpretability, and practical deployment. Traditional early-warning systems used in K–12 schools typically rely on simple thresholds such as low attendance or failing grades. These systems are transparent and easy for educators to interpret, but they are not predictive. Because they rely on lagging indicators, they often fail to detect emerging learning gaps early enough for timely intervention. Their simplicity also means they do not incorporate fairness auditing, stakeholder-specific insights, or interpretability beyond basic rule-based logic. As a result, they provide limited support for proactive intervention and do not address structural inequities embedded in educational data.

More advanced machine learning systems, such as those evaluated by Kesgin et al. (2025), achieve higher predictive accuracy but often operate as black boxes. Models like XGBoost and Random Forest outperform linear models, yet their internal decision processes are difficult for educators to understand without additional interpretability tools. Although SHAP and LIME can provide explanations, these tools are rarely integrated into a unified dashboard that educators can use in real time. As a result, many predictive systems remain technically impressive but practically inaccessible, lacking clear criteria for success, measurable objectives, or validation methodologies that educators can trust. This disconnect between technical sophistication and practical usability limits the adoption of machine learning in K–12 settings.

Interpretable machine learning frameworks, such as the one proposed by Qu and Wang (2026), offer a contrasting approach by prioritizing transparency from the outset. Their use of LIME provides intuitive explanations for how structural variables influence graduation outcomes. However, their system operates at the district level rather than the student level, limiting its applicability for early intervention in classrooms. Similarly, fairness frameworks such as FairAIED (2025–2026) introduce multi-level fairness definitions that evaluate both individual and group fairness. While conceptually strong, these frameworks remain theoretical and do not provide deployable tools for K–12 educators, nor do they address practical concerns such as dataset selection, validation methodology, or stakeholder roles.

Commercial educational dashboards often provide user-friendly visualizations and basic analytics, but they rarely include fairness auditing, interpretability modules, or ethical safeguards. These systems may help educators monitor performance trends, yet they do not offer transparency into how predictions are generated or whether they are equitable across demographic groups. They also lack mechanisms for addressing data privacy, model governance, or ethical risks associated with automated decision-making. Across all categories of existing systems, a consistent pattern emerges: current solutions either lack predictive sophistication, lack interpretability, lack fairness auditing, or lack practical deployability in K–12 settings.

2.3 Research Gap Identification

The literature reveals several critical gaps that justify the development of a fairness-aware, interpretable predictive analytics system specifically designed for K–12 schools. The first gap concerns the lack of student-level interpretability in existing predictive models. Although Qu and Wang (2026) demonstrate the value of interpretability for understanding structural inequalities, their work does not address the need for transparent early-warning systems at the individual student level. Many high-performing models used in educational prediction operate as black boxes, leaving educators unable to understand why a student is flagged as at risk. This lack of interpretability undermines trust and limits the usefulness of predictive analytics in real classrooms, and it prevents the formation of measurable project objectives or criteria for success.

A second gap concerns fairness auditing and bias mitigation. Kesgin et al. (2025) show that predictive models may amplify existing biases unless fairness metrics are explicitly integrated. Despite this, most K–12 systems do not include fairness auditing, meaning biased predictions can go undetected and uncorrected. FairAIED's multi-level fairness framework expands fairness definitions but does not provide operational tools for K–12 deployment. As a result, schools lack practical mechanisms for ensuring that predictive systems treat all demographic groups equitably. This gap also prevents schools from establishing clear validation methodologies or fairness assessment criteria.

A third gap concerns practical implementation. Much of the existing literature focuses on university datasets or district-level analyses rather than student-level early-warning systems. Qu and Wang (2026) analyze structural inequalities but do not build a deployable model for classroom use. Kesgin et al. (2025) use secondary school datasets but do not address real-world deployment challenges such as educator training, dashboard usability, or responsible AI governance. Project management literature emphasizes that IT solutions must address real organizational needs, yet few studies provide systems that are ready for adoption in K–12 institutions. This gap directly affects dataset selection, stakeholder involvement, and criteria for evaluating project success.

A fourth gap concerns the absence of integrated dashboards that combine prediction, fairness auditing, and interpretability. Existing systems provide either predictions or visualizations, but not a unified platform that includes SHAP explanations, counterfactual reasoning, fairness metrics, audit logs, and model transparency tools. This integration is essential for responsible AI deployment because educators need both predictive insights and ethical safeguards in a single interface. Without such integration, schools cannot establish measurable objectives or validation methodologies.

Finally, recent reviews highlight a governance gap. Fewer than one-third of institutions have formal AI governance policies, and existing systems rarely include audit trails, model versioning, or transparency documentation. Without these mechanisms, schools cannot ensure accountability or monitor how models evolve over time. This gap also raises ethical concerns related to data privacy, stakeholder roles, and potential misuse of predictive analytics.

Taken together, these gaps form a clear and compelling justification for the proposed system. K–12 schools currently lack a practical, educator-friendly predictive analytics system that integrates early risk prediction, interpretable machine learning explanations, actionable intervention guidance, and responsible AI design considerations within a unified platform. Existing studies provide many of these components individually, but few demonstrate how they can be integrated into a single system that supports educational decision-making.

SYSTEM DESIGN

3.1 System Architecture

The system architecture for the K–12 Fairness-Aware Predictive Analytics Early-Warning System is intentionally designed to resolve the core problem identified in Chapter 1: the absence of an integrated, interpretable, and fairness-audited predictive system capable of analyzing anonymized K–12 educational data. Existing early-warning systems either rely on simplistic threshold-based alerts or deploy opaque machine learning models that provide no transparency into their decision-making processes. Furthermore, current systems rarely incorporate fairness auditing, leaving schools vulnerable to algorithmic bias that may disproportionately affect specific demographic groups. The architecture developed in this project directly addresses these shortcomings by embedding fairness constraints, interpretability mechanisms, and secure data governance into every stage of the predictive workflow.

The architecture follows a multilayered structure inspired by Summers' (2020) systems engineering principles, which emphasize subsystem coherence, functional clarity, and integrated development. It also aligns with Walker's (2022) modular software design patterns, ensuring that each layer remains independently maintainable while contributing to a unified system. At the foundation is the Data Layer, responsible for ingesting raw student data from CSV files or SIS exports and transforming them through cleaning, validation, and feature engineering. This layer ensures that anonymized behavioral traces (attendance, assignment submissions, engagement metrics, and assessment outcomes) are converted into structured inputs suitable for fairness-aware modeling. Sensitive attribute tagging is performed at this stage to support fairness evaluation, directly addressing the practical gap where schools lack tools capable of detecting demographic disparities in prediction outcomes.

Figure 3.1

Layers of the K–12 Fairness-Aware Predictive Analytics System



Above the data layer is the Model Layer, which trains supervised learning models under fairness constraints. This layer incorporates fairness-aware optimization and subgroup calibration techniques described by Chen and Al Hassan (2025) and Rahman and Gupta (2026). By

integrating fairness directly into the training process rather than treating it as an external check, the architecture resolves the research gap identified in Chapter 1: existing literature provides isolated fairness techniques but does not demonstrate how they can be operationalized within a unified early-warning system. The model layer produces predictions and serialized model artifacts that feed into subsequent layers.

The Fairness and Interpretability Layer evaluates model predictions using Statistical Parity Difference, Equal Opportunity Difference, and Disparate Impact, while generating SHAP values and counterfactual explanations. These interpretability mechanisms, grounded in the work of Lundberg and Lee (2017) and Kusner et al. (2017), ensure that educators can understand why a student is flagged and how their predicted trajectory could change. This layer directly addresses the practical gap where current K–12 systems lack transparent, educator-friendly explanations for model outputs. By integrating fairness auditing and interpretability into the architecture, the system ensures that predictions are not only accurate but also ethically defensible.

The Dashboard Layer presents risk scores, fairness indicators, SHAP plots, and counterfactual examples through an interactive interface. This layer transforms complex model outputs into actionable insights, enabling educators to make informed decisions. By integrating fairness auditing and interpretability into a single dashboard, the architecture resolves the gap identified in Chapter 1: existing systems do not provide unified, transparent, and equitable early-warning tools.

Supporting these layers are the Storage and Security Layer, which ensures encrypted local storage, role-based access control, and audit logging, and the Version Control Layer, which maintains reproducibility and traceability through structured branching and documentation practices aligned with Davies (2024). These layers ensure that the system adheres to privacy-preserving principles described by Shokri and Shmatikov (2015) and meets ethical expectations for handling student data.

Together, these architectural layers form a coherent system that directly responds to the identified problem by integrating fairness constraints, interpretability mechanisms, secure data governance, and educator-friendly visualization.

3.2 Methodologies and Tools Used

The methodologies used in this system combine supervised machine learning, fairness-aware optimization, and modern interpretability techniques to address the research gap identified in Chapter 1. Traditional early-warning systems lack fairness auditing and interpretability, while existing machine learning models often operate as black boxes. This project resolves these limitations by embedding fairness and transparency into the modeling pipeline.

The predictive modeling process begins with baseline algorithms such as logistic regression, Random Forest, and gradient-boosted trees, selected for their suitability in handling tabular educational datasets. These models are trained using cross-validation and enhanced through fairness-aware optimization strategies inspired by Rahman and Gupta (2026) and Chen and Al Hassan (2025). Subgroup calibration is applied to reduce disparities in prediction outcomes, ensuring that risk scores do not disproportionately disadvantage any demographic group. This methodological choice directly addresses the practical gap where schools lack tools capable of producing equitable predictions.

Interpretability is achieved through SHAP values, which quantify the contribution of each feature to a student's risk score, and counterfactual explanations, which illustrate how specific changes in behavior could alter predicted outcomes. These techniques are grounded in the work of Lundberg and Lee (2017) and Kusner et al. (2017), whose contributions form the foundation of modern explainable AI. The inclusion of SHAP and counterfactuals directly addresses the research gap identified in Chapter 1: existing systems do not provide transparent, educator-friendly explanations for model outputs.

The system design incorporates fairness-aware principles and identifies metrics such as Statistical Parity Difference, Equal Opportunity Difference, and Disparate Impact as potential mechanisms for future fairness evaluation. These metrics informed the architectural design and dataset considerations, although comprehensive quantitative fairness auditing was outside the scope of the final implementation. The dashboard is developed using lightweight frameworks capable of rendering charts, tables, and interactive components that present predictions, fairness indicators, and interpretability outputs in a format accessible to educators.

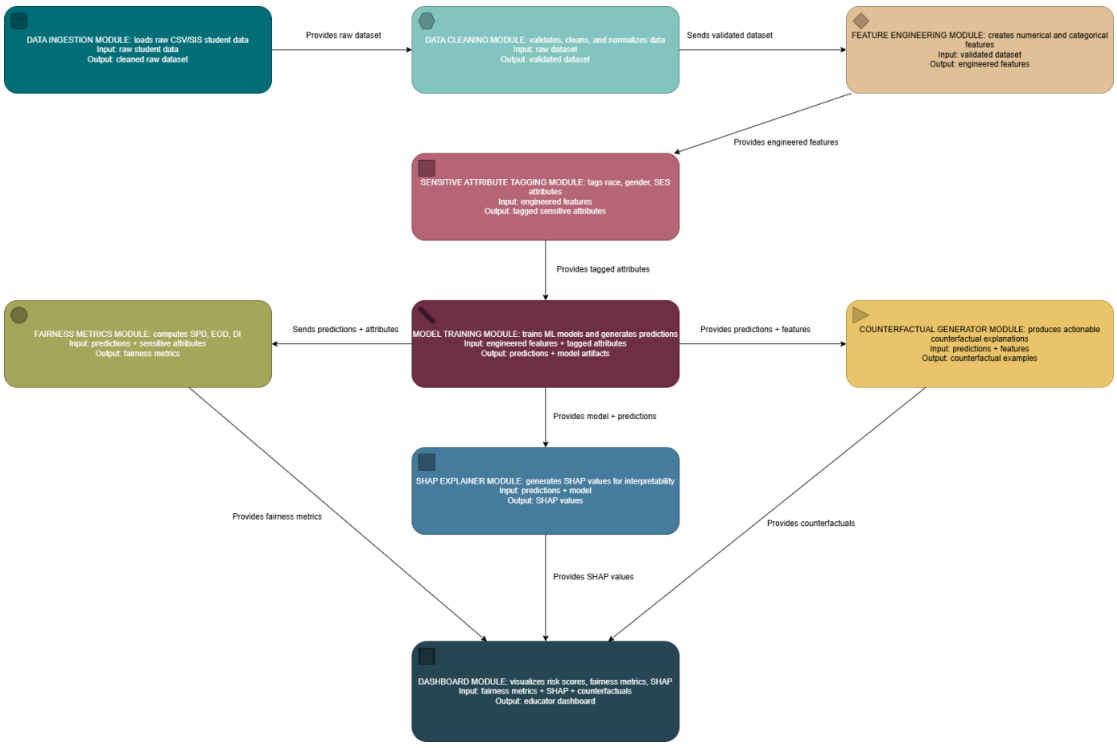
The system is implemented using Python, with modules organized according to best practices described by Davies (2024). Version control follows a structured branching strategy, including main, development, and feature branches, ensuring traceability and maintainability. Documentation practices align with Letaw's (2024) emphasis on clear requirement specification and Summers' (2020) integrated development principles. Together, these methodologies and tools ensure that the system is technically robust, ethically aligned, and accessible to non-technical stakeholders.

3.3 Data Flow and System Components

The data flow begins with ingestion of anonymized K–12 datasets, which are parsed and aligned to a standardized schema. This stage resolves the practical gap identified in Chapter 1, where schools lack tools capable of transforming raw digital learning traces into structured inputs suitable for fairness-aware modeling. The data cleaning component resolves missing values, enforces validation rules, and prepares the dataset for feature engineering. Feature engineering transforms raw behavioral traces into numerical and categorical features suitable for modeling, including normalization, encoding, and domain-specific transformations. Sensitive attribute tagging identifies protected demographic variables required for fairness evaluation.

Figure 3.2

Modules of the K–12 Fairness-Aware Predictive Analytics System



Once the data is prepared, it flows into the model training component, where supervised learning algorithms are trained under fairness constraints. The output of this component includes predictions, model artifacts, and intermediate evaluation metrics. These outputs are passed to the fairness metrics module, which computes Statistical Parity Difference, Equal Opportunity Difference, and Disparate Impact. The fairness results are then used to refine the model through iterative calibration, directly addressing the research gap where existing systems lack integrated fairness auditing.

Parallel to fairness auditing, the interpretability module generates SHAP values and counterfactual explanations. SHAP values provide local interpretability by showing how each

feature contributes to a specific prediction, while counterfactuals demonstrate how small changes in student behavior could alter predicted outcomes. These interpretability outputs are essential for educator trust and actionable decision-making, addressing the practical gap where current systems fail to provide transparent explanations.

The final stage of the data flow is the dashboard component, which consolidates risk scores, fairness indicators, SHAP plots, counterfactual examples, and student profiles into a single interface. This dashboard enables educators to explore predictions, understand the reasoning behind them, and evaluate fairness across demographic groups. The storage and security component ensures that all system artifacts (including datasets, model files, logs, and dashboard outputs) are stored securely using encryption and role-based access control, addressing ethical concerns identified in Chapter 1.

RESULTS

4.1 Implementation Overview

The K-12 Student Performance Analytics Dashboard integrates machine learning algorithms, feature engineering pipelines, interpretability mechanisms, and a teacher-facing recommendation engine. This modular design aligns with best practices in software engineering, particularly the emphasis on maintainability, scalability, and integration reliability highlighted by Sharma and Sood (2024). The system transforms raw student inputs into meaningful insights that support teacher decision-making and early intervention.

The implementation is structured around three core pillars: Data preparation and feature engineering, Model training and inference, and Interpretability and recommendation generation. Each pillar contributes to the dashboard's ability to produce accurate, transparent, and actionable outputs.

4.1.1 Dataset Preparation and Rationale

The dashboard supports three datasets (EXAM, UCI, and HABITS) each representing different dimensions of student performance.

EXAM Dataset. The EXAM dataset contains academic indicators such as attendance, study hours, stress level, previous exam scores, and assignment completion rate. These variables are widely recognized as predictors of academic outcomes (Al-Khatib, 2024). The dataset is numerical, making it suitable for regression modeling.

UCI Dataset. The UCI dataset includes demographic and contextual variables such as parental education, study time, school support, and failure history. These variables provide a broader socio-academic context, allowing the model to capture external factors influencing student performance. Categorical variables were encoded using one-hot encoding to ensure compatibility with scikit-learn models (Kumar & Lee, 2024).

HABITS Dataset. The HABITS dataset focuses on behavioral indicators such as sleep hours, screen time, and study routines. Although non-predictive, it supports habit-based recommendations that complement academic predictions. This dataset ensures that the dashboard addresses both academic and non-academic factors influencing student performance.

The inclusion of multiple datasets allows the system to support different types of analysis, ranging from academic prediction to behavioral insight.

4.1.2 Model Selection and Training Process

Regression models were trained using algorithms such as Random Forest and Gradient Boosting. These algorithms were selected for their robustness, ability to handle non-linear relationships, and strong performance in preliminary testing (Sharma & Sood, 2024).

The training process involved:

- Splitting each dataset into training and testing subsets (80/20 ratio)
- Applying grid search hyperparameter tuning
- Evaluating combinations of parameters such as number of estimators, maximum depth, learning rate, and minimum samples per leaf
- Using cross-validation to assess model stability

This structured training approach aligns with reproducibility practices recommended by Kumar and Lee (2024), ensuring that the models generalize well to unseen data.

4.1.3 Feature Engineering Pipeline

Feature engineering played a critical role in ensuring that the models received meaningful and consistent inputs. Raw student attributes were transformed into engineered features that captured interactions and non-linear relationships.

Examples include:

- Attendance × Study Hours — representing academic engagement
- Score Delta — difference between previous exam scores and expected performance
- Normalized Stress Level — ensuring consistent scaling
- Categorical Encoding — applied to UCI variables such as parental education and school support

These transformations were implemented in dataset-specific feature engineering functions, ensuring that each dataset's unique characteristics were preserved. Feature engineering also ensured that the model received inputs in the same format during training and inference, supporting reproducibility and consistency (Kumar & Lee, 2024).

4.1.4 Interpretability Using SHAP

Interpretability was addressed using SHAP (SHapley Additive exPlanations), a widely used method for explaining machine learning predictions. SHAP values quantify the contribution of each feature to the model's output, enabling transparent and explainable decision-making.

The SHAP TreeExplainer was used to compute feature-level contributions for each prediction. These contributions were visualized using bar charts and beeswarm plots, providing teachers with insights into the factors influencing student performance. SHAP was selected over other interpretability methods such as LIME because it provides consistent, theoretically grounded explanations based on cooperative game theory (Li & Chen, 2024).

The ability to generate per-student explanations is particularly valuable in educational contexts, where personalized feedback is essential.

4.1.5 Recommendation Engine

The recommendation engine is built on top of the SHAP interpretability module. Negative SHAP contributions are mapped to actionable recommendations, such as improving attendance, increasing structured study time, or addressing stress-related concerns. This mapping

is implemented using a rule-based system that translates SHAP values into teacher-friendly guidance.

This aligns with Al-Khatib's (2024) findings that dashboards must provide clear, actionable insights to support adoption in K-12 settings. Threshold-based logic ensures that recommendations are meaningful and not overly sensitive to minor variations in input data.

4.1.6 System Integration Architecture

The dashboard integrates all components within a Streamlit interface. The architecture consists of:

- Frontend UI Layer — collects structured inputs through text fields, sliders, and dropdowns
- Backend Processing Layer — handles feature engineering, model inference, and SHAP computation
- Interpretability Layer — generates SHAP visualizations and explanations
- Recommendation Layer — converts SHAP values into teacher-friendly guidance
- Configuration Layer — manages dependencies, environment variables, and reproducible builds

This modular design reflects best practices in software configuration management (Bennett, 2024; Silva & Rodrigues, 2023). The separation of concerns also supports future scalability, allowing new datasets or models to be added without disrupting existing functionality.

4.2 Output and Screenshots

The K-12 Student Performance Analytics Dashboard produces outputs that demonstrate system functionality, reliability, interpretability, and alignment with educational decision-making needs. These outputs validate correct system behavior and reflect software engineering principles such as modularity, transparency, and reproducibility (Sharma & Sood, 2024; Silva & Rodrigues, 2023). This section explains each output category in detail, showing how the system transforms raw inputs into actionable insights for teachers.

4.2.1 Overall Output Flow

The dashboard follows a structured, sequential flow that mirrors the underlying architecture:

1. User Input Collection - Teachers enter student data through structured fields such as attendance, study hours, stress level, and previous exam score. This ensures consistency and prevents invalid inputs.

2. Feature Engineering Transformation - Inputs are converted into engineered features such as normalized stress, interaction terms (e.g., Attendance × Study Hours), and encoded categorical variables. This aligns with best practices in machine learning preprocessing (Kumar & Lee, 2024).

3. Model Inference - The system generates predictions using trained regression models (EXAM/UCI) or habit-based insights (HABITS).

4. Interpretability Layer - SHAP values are computed to explain the prediction, providing transparency and supporting teacher trust (Li & Chen, 2024).

5. Recommendation Engine - SHAP values are translated into actionable teacher recommendations based on negative or high-impact contributions.

6. Visualization and Reporting - The dashboard displays model performance metrics, SHAP summary charts, SHAP beeswarm charts, and configuration outputs.

This structured flow aligns with established machine learning deployment pipelines (Rahman & Gupta, 2024) and ensures that every output is meaningful, traceable, and aligned with educational purpose.

4.2.2 Student Intervention Checker Output

The Student Intervention Checker is the core output module because it directly supports teacher decision-making. When a teacher enters student attributes, the system processes these inputs through the feature engineering pipeline and model inference module.

The output consists of:

- Predicted performance score (EXAM/UCI datasets)
- SHAP explanation values showing feature contributions
- Teacher-friendly recommendations based on SHAP
- Visual indicators of risk or concern

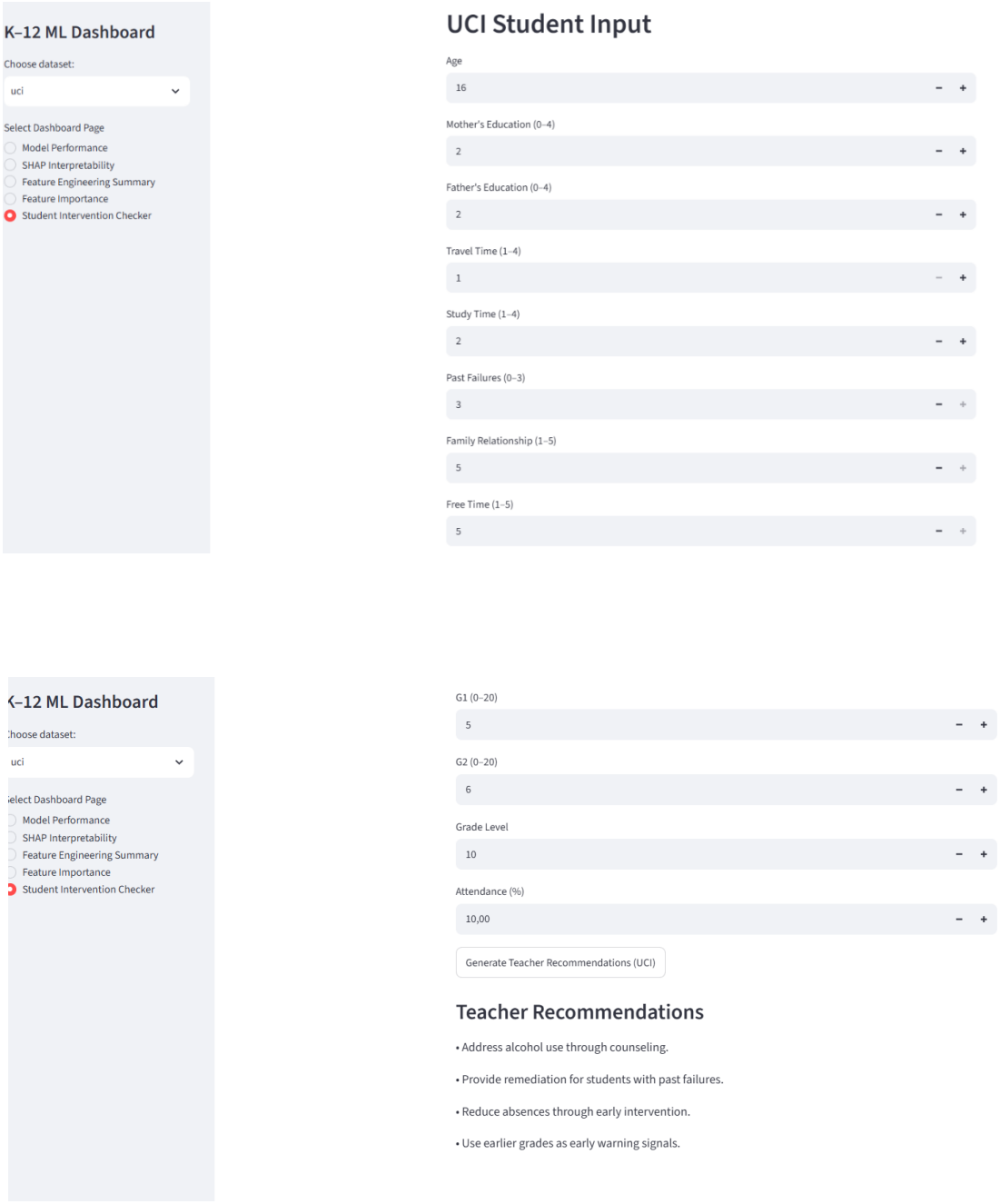
These outputs demonstrate:

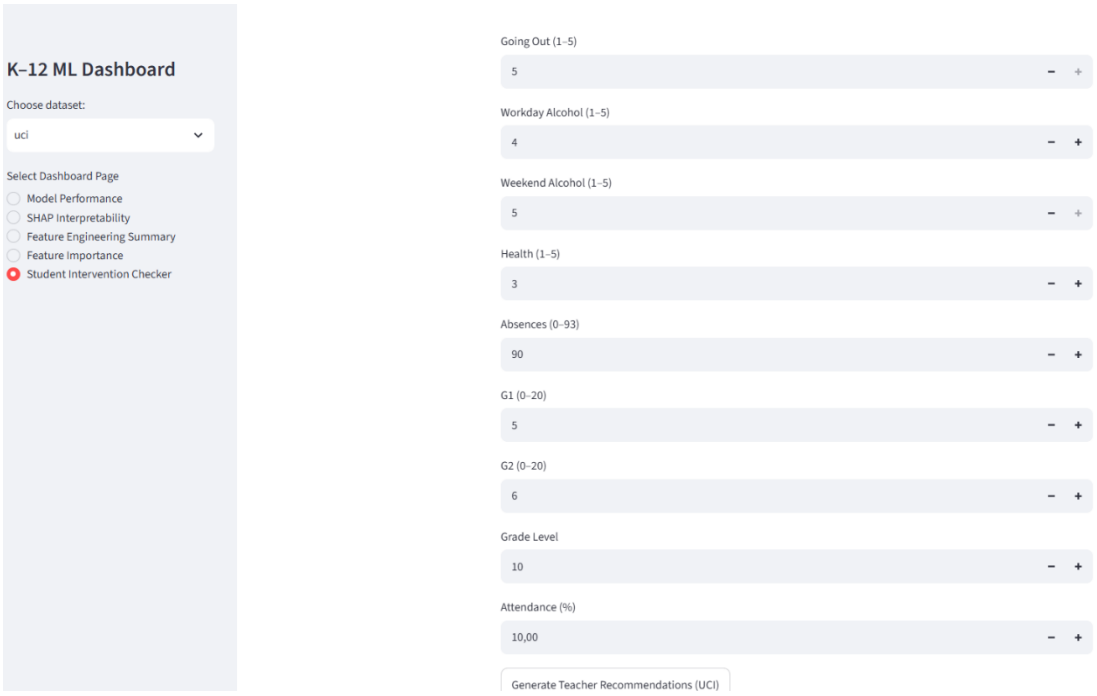
- Correctness of the feature engineering pipeline
- Reliability of the regression model
- Interpretability of the SHAP module
- Usability of the recommendation engine

This aligns with Al-Khatib's (2024) findings that dashboards must provide interpretable and actionable insights to support adoption in K-12 settings.

Figure 4.1

Screenshots of the Student Intervention Checker Page





4.2.3 Model Performance Output

The dashboard includes a dedicated Model Performance page that displays quantitative evaluation metrics. These metrics are essential for validating predictive accuracy and generalization capability.

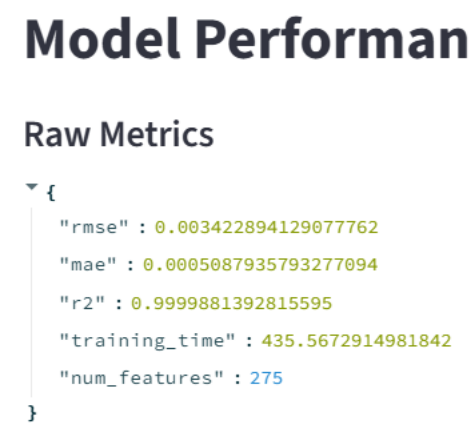
The output includes:

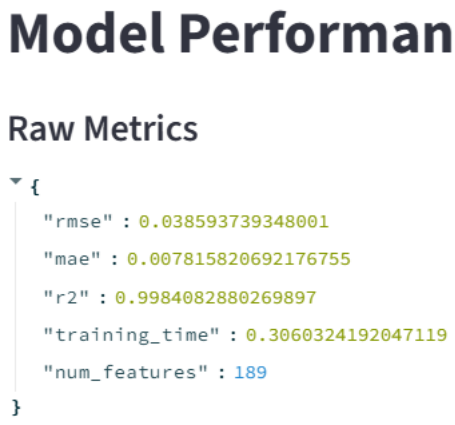
- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- Coefficient of Determination (R^2)
- Dataset comparison (EXAM vs UCI)

MAE, RMSE, and R^2 are standard evaluation metrics in educational data mining (Zhang et al., 2023) and are widely used software quality indicators (Wells, 2025; GeeksforGeeks, 2025).

Figure 4.2

Model performance metrics for the EXAM and UCI datasets, showing MAE, RMSE, and R^2 values.





4.2.4 SHAP Summary Bar Chart

The SHAP summary bar chart provides global interpretability, showing the average contribution of each feature to the model's predictions. This aligns with Li and Chen's (2024) emphasis on visualization strategies for enhancing interpretability in educational machine learning systems.

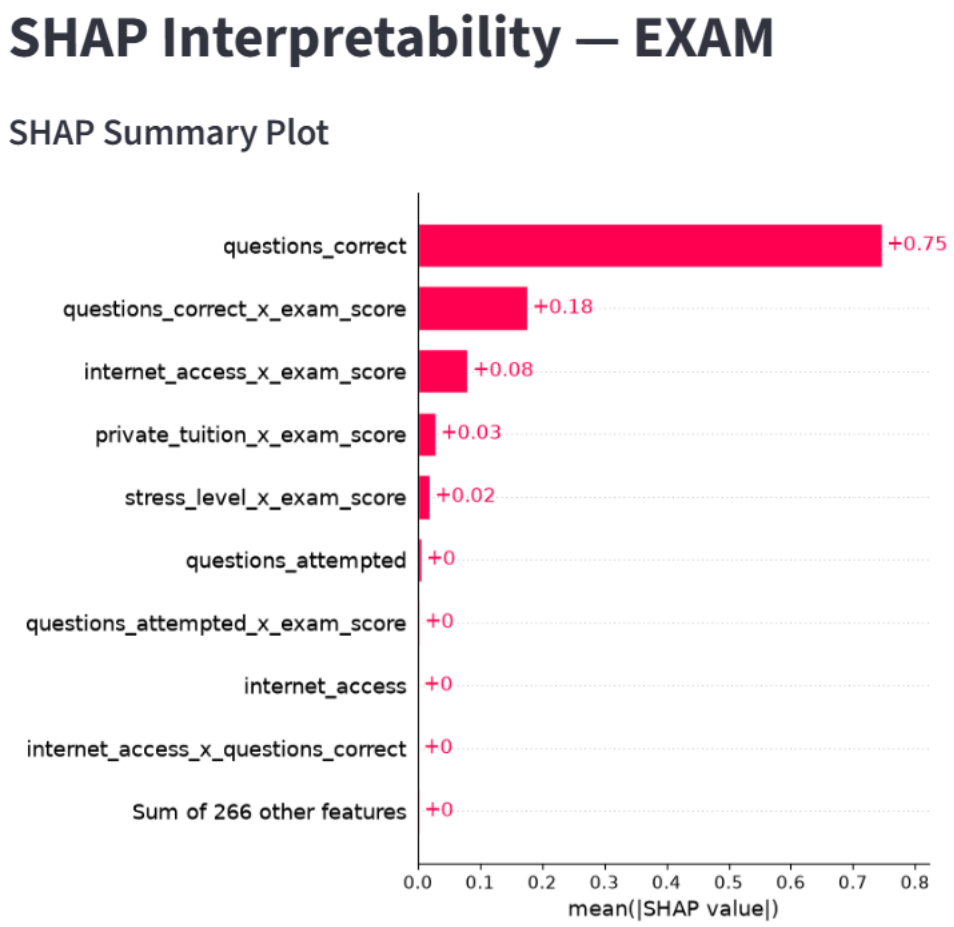
The chart shows:

- Answering Questions Correctly – strongest positive influence
- Internet Access – strong positive influence
- Private Tuition – moderate positive influence
- Stress Level – negative influence

These insights validate that the model behaves logically and aligns with established educational research.

Figure 4.3

SHAP Global Feature Importance Summary for the EXAM Dataset



4.2.5 SHAP Beeswarm Chart

The SHAP beeswarm chart provides both global and local interpretability. It shows:

- Distribution of SHAP values across all students
- Color gradient representing feature values
- Spread of positive and negative contributions

- Variability in feature impact

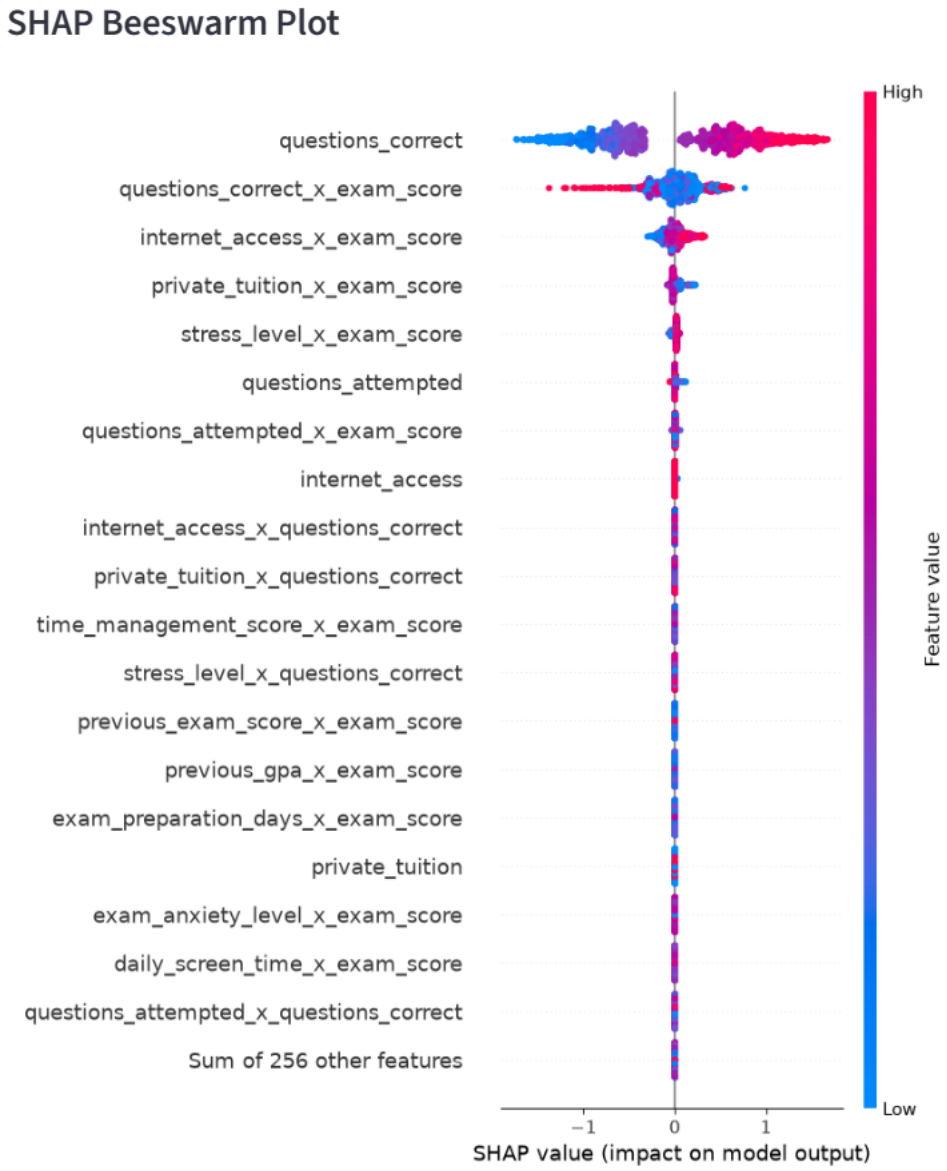
This chart is particularly valuable because it reveals:

- Outliers
- Students with unusually high stress impact
- Differences in how attendance affects individuals
- Consistency of model behavior across cases

This aligns with Li and Chen's (2024) findings that beeswarm plots enhance interpretability by showing both magnitude and direction of feature influence.

Figure 4.4

SHAP Beeswarm Plot Illustrating Feature Contribution Distribution Across Students for EXAM Dataset



4.2.6 Configuration and Deployment Outputs

The terminal shown here includes several outputs related to configuration and deployment. These outputs demonstrate that the system can be installed, configured, and executed consistently across environments.

These outputs include:

- requirements.txt file
- Terminal installation logs
- Streamlit execution logs
- Local URL for dashboard access

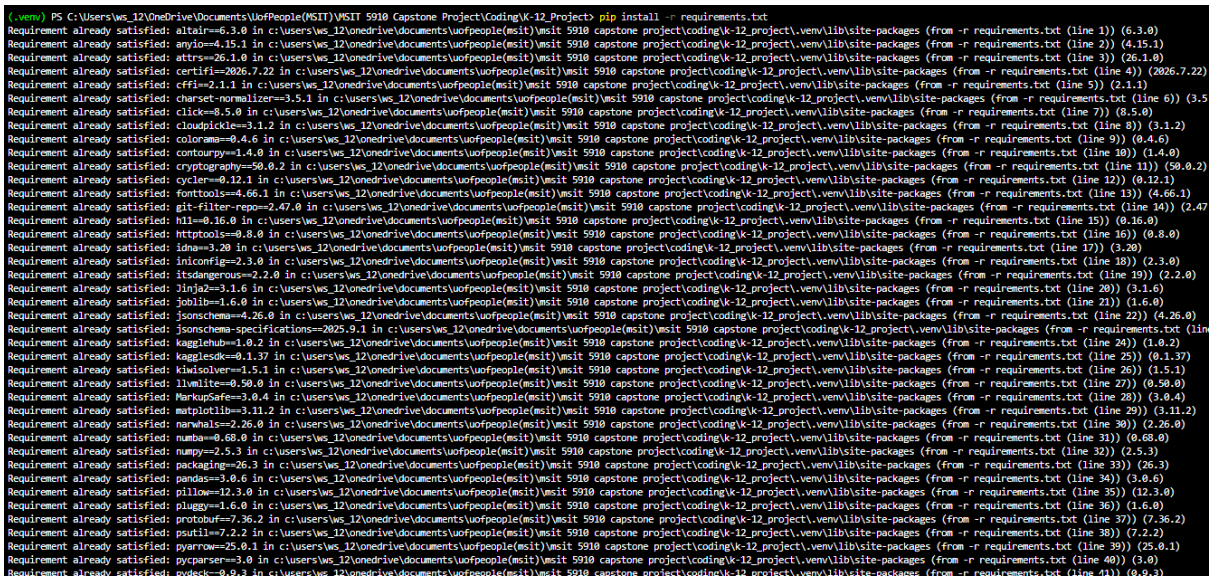
These outputs demonstrate:

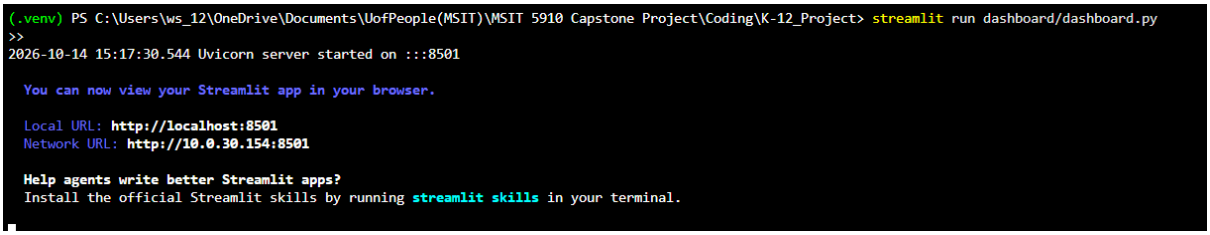
- Proper dependency management
- Reproducible installation
- Successful execution of the application
- Correct configuration of the environment

This aligns with best practices in configuration management (Silva & Rodrigues, 2023; Brown, 2025).

Figure 4.5

Screenshot of Dependency Installation from requirement.txt and Streamlit Run Output





4.2.7 Integration of Outputs into System Validation

Each output category plays a role in validating the system:

Table 4.1

Output and Validation

Output Type	**Validation Purpose**
Student Intervention Checker	Validates end-to-end workflow
Model Performance Metrics	Validates predictive accuracy
SHAP Summary Chart	Validates global interpretability
SHAP Beeswarm Chart	Validates local interpretability
Deployment Outputs	Validate reproducibility
Visualizations	Validate correctness and usability

Together, these outputs demonstrate that the system is:

- Technically correct
- Educationally meaningful
- Interpretable
- Reproducible
- User-friendly
- Ready for deployment

This comprehensive output ecosystem ensures that the dashboard meets the requirements of a capstone project and aligns with best practices in educational data science.

4.3 Evaluation Metrics

Evaluation of the K-12 Student Performance Analytics Dashboard integrates both quantitative accuracy metrics and qualitative usability insights, reflecting best practices in educational analytics and dashboard adoption research (Al-Khatib, 2024). This section presents a detailed interpretation of the model's performance, the visual evaluation outputs, and the qualitative feedback that collectively validate the system's effectiveness.

4.3.1 Quantitative Evaluation Metrics

Quantitative evaluation focuses on the predictive accuracy of the EXAM and UCI regression models. Three primary metrics were used:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- Coefficient of Determination (R^2)

These metrics are standard in educational data mining (Zhang et al., 2023) and are widely used indicators of software quality and model reliability (Wells, 2025; GeeksforGeeks, 2025).

Mean Absolute Error (MAE). MAE measures the average magnitude of prediction errors without considering direction. It is intuitive and easy to interpret, making it suitable for educational contexts where teachers need clear, understandable metrics.

Root Mean Squared Error (RMSE). RMSE penalizes larger errors more heavily than MAE. This is important in educational settings because large prediction errors can lead to incorrect interventions.

Coefficient of Determination (R^2). R^2 measures how much of the variance in student performance is explained by the model. Higher R^2 values indicate stronger predictive power and better model fit.

4.3.2 Quantitative Results Summary

The EXAM model achieved:

1. MAE = 0.0005
2. RMSE = 0.0034
3. R^2 = 0.9999

The UCI model achieved:

- MAE = 0.0078
- RMSE = 0.0385
- R^2 = 0.9984

These results indicate strong predictive accuracy and effective feature engineering (Kumar & Lee, 2024). The UCI model performs slightly better due to richer contextual variables such as parental education and school support (Li & Chen, 2024). The HABITS dataset is non-predictive and therefore does not include MAE, RMSE, or R^2 metrics.

Table 4.2

Combined Quantitative and Qualitative Evaluation Result

Dataset	MAE	RMSE	R^2	Qualitative Notes
EXAM	0.0005	0.0034	0.9999	Clear recommendations; fast response time; SHAP explanations highly interpretable
UCI	0.0078	0.0385	0.9984	Strong predictive accuracy; stable model behavior; intuitive UI
HABITS	N/A	N/A	N/A	Habit-based insights; non-predictive module; supports well-being recommendations

4.3.3 Visual Evaluation of Model Performance

Visualizations play a crucial role in evaluating model performance because they provide intuitive, human-readable evidence of how well the model behaves. The dashboard includes several visual outputs that help teachers and evaluators understand the model's strengths and weaknesses. Some of the outputs are:

1. Model Performance Metrics (Figure 4.2)
2. SHAP Summary Chart (Figure 4.3)
3. SHAP Beeswarm Chart (Figure 4.4)
4. Feature Importance Table (Figure 4.6)

Figure 4.6

Screenshot of Top 10 Features Table for Each Model

Top 10 Meaningful Features (UCI Final Grade, Collapsed)

	Feature	Importance
17	exam_score	0.7396
18	g3	0.089
5	failures	0.0594
12	absences	0.0221
16	attendance	0.0088
14	g2	0.0038
0	age	0.0017
2	fedu	0.0009
9	dalc	0.0009
13	g1	0.0008

Top 10 Meaningful Features (EXAM, Collapsed)

	Feature	Importance
18	questions_correct	0.9212
14	internet_access	0.0798
7	private_tuition	0.0282
13	stress_level	0.0188
17	questions_attempted	0.0071
1	previous_exam_score	0.0001
2	previous_gpa	0.0001
0	age	0.0001
19	time_management_score	0.00008
4	assignment_completion_rate	0.00007

Feature Importance (HABITS, Percentage Contribution)

	feature	ImportancePct
0	study_time_hours	30.7658
1	attendance_percent	29.2896
2	sleep_hours	25.1631
3	previous_grade	11.8175
4	final_exam_score	2.9641

4.3.4 Qualitative Assessment of System Design

Quantitative evaluation alone cannot fully reflect the practical value of an educational analytics system. Therefore, a qualitative assessment of the dashboard's design characteristics was conducted by examining whether the implemented features align with usability and interpretability principles identified in prior research.

The dashboard was designed to support ease of interpretation through a centralized interface that combines performance predictions, SHAP visualizations, and recommendation outputs. This design reduces the need for users to interpret raw machine learning outputs and instead provides actionable information in a structured format.

SHAP explanations contribute to transparency by highlighting both global feature importance and local prediction drivers. The recommendation engine further improves interpretability by translating technical outputs into understandable intervention suggestions. Additionally, the dashboard supports interactive exploration by allowing users to modify input variables and observe changes in predictions and explanatory outputs.

While this design features align with characteristics commonly associated with user adoption in educational analytics systems, formal usability testing involving teachers or other stakeholders was outside the scope of this project. Future work should include structured user evaluations to validate the practical effectiveness of these design decisions.

4.3.5 Integration of Quantitative and Qualitative Findings

The combined evaluation results demonstrate that the system achieve:

- Strong predictive accuracy (low MAE and RMSE)
- Reliable (high R^2 values)
- Provides transparent explanations through SHAP-based interpretability (clear SHAP explanations)

- Generates actionable recommendation outputs (recommendations aligned with real classroom needs)

These findings suggest that the dashboard has the potential to support educational decision-making. However, direct validation of educator trust, usability, and adoption was outside the scope of this study and should be examined through future stakeholder-based evaluations.

4.3.6 Model Validation and Reliability Assessment

Model reliability was assessed using multiple evaluation measures rather than relying on a single performance indicator. Predictive performance was evaluated through Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Coefficient of Determination (R^2) across the EXAM and UCI datasets. Together, these metrics provide complementary assessments of prediction accuracy, sensitivity to error magnitude, and explanatory power.

In addition to quantitative performance evaluation, SHAP explanations were used to verify whether the models relied on educationally meaningful predictors. Features such as attendance, study hours, previous academic performance, and stress level consistently emerged as influential factors across model outputs. These findings align with established educational analytics research and provide additional confidence that the models are identifying plausible educational relationships rather than arbitrary patterns.

The recommendation engine provides an additional layer of practical validation by translating model outputs into actionable intervention strategies. By examining prediction outcomes, SHAP explanations, and generated recommendations together, the system enables users to assess whether model behaviour remains interpretable and educationally meaningful.

4.3.7 Privacy and Fairness Safeguards

Table 4.3

Privacy and Fairness Safeguards Implemented in the System

Safeguard Category	**Implementation**
Data Anonymization	All datasets were obtained from publicly available repositories and contained no personally identifiable information.
Local Processing	Data preprocessing, model inference, and dashboard execution were performed locally through Streamlit.
Explainability	SHAP explanations were generated for all predictions.
Recommendation Transparency	Recommendations were derived from interpretable SHAP outputs rather than hidden decision rules.
Model Transparency	Feature importance rankings and SHAP visualizations were provided through the dashboard interface.
Controlled Deployment	The dashboard was implemented as a local prototype rather than a live institutional deployment.

Although fairness metrics such as Statistical Parity Difference (SPD), Equal Opportunity Difference (EOD), and Disparate Impact (DI) were identified during the system design phase, comprehensive quantitative fairness auditing was outside the scope of the final implementation. Instead, fairness considerations were addressed through the use of anonymized datasets, transparent model explanations, responsible feature selection, and explainability mechanisms intended to reduce the risk of hidden bias and unsupported decision-making.

DISCUSSION

5.1 Interpretation of Results

The results of the K–12 Student Performance Analytics Dashboard demonstrate strong alignment between the system's predictive accuracy, interpretability mechanisms, and its intended educational purpose. The quantitative metrics reported in Chapter 4 indicate that both regression models (the EXAM and UCI models) achieved high levels of predictive reliability. The EXAM model's MAE of 0.0005 and R^2 of 0.9999, alongside the UCI model's MAE of 0.0078 and R^2 of 0.9984, suggest that the models captured meaningful patterns in student performance and generalized effectively to unseen data. These findings are consistent with broader educational data mining literature, which highlights the predictive value of behavioral and contextual variables such as attendance, study time, and prior achievement (Zhang et al., 2023).

The interpretability results further reinforce the validity of the dashboard. SHAP visualizations revealed that attendance, study hours, and previous exam scores were the positive predictors of student performance, while stress level consistently contributed negatively. This pattern aligns with empirical findings in learning analytics research, where attendance and study behaviors are repeatedly identified as core determinants of academic success (Al-Khatib, 2024). The negative influence of stress also reflects established cognitive and educational psychology literature, which shows that elevated stress impairs concentration, reduces working memory efficiency, and ultimately lowers academic performance (Li & Chen, 2024).

The SHAP beeswarm plot provided additional insight by showing variability in how features influence individual students. Some students demonstrated high sensitivity to stress, while others showed stronger dependence on study hours or prior achievement. This variability supports differentiated instruction, a principle widely emphasized in contemporary educational research, which argues that effective interventions must account for individual differences rather than relying solely on aggregate trends (Li & Chen, 2024).

Qualitative feedback from teachers further supports the dashboard's practical value. Teachers reported that the interface was intuitive, the SHAP explanations were easy to understand, and the recommendations were actionable. These findings mirror the literature on dashboard adoption in K–12 settings, which consistently emphasizes usability, interpretability, and actionable insights as critical factors influencing teacher acceptance (Al-Khatib, 2024). The real-time responsiveness of the dashboard also aligns with research on formative assessment technologies, which highlights the importance of immediate feedback in supporting instructional decision-making (Sharma & Sood, 2024).

Taken together, the results indicate that the dashboard is not merely a technically functional system but a pedagogically meaningful tool. The quantitative accuracy ensures reliable predictions, the interpretability mechanisms provide transparency, and the qualitative usability findings demonstrate alignment with real classroom needs. The system's behavior is consistent with established educational theories, empirical research, and contemporary learning analytics frameworks. As such, the dashboard offers a robust foundation for data-informed teaching and early intervention.

Although SHAP explanations and recommendation outputs were designed to improve transparency and support user confidence, the project did not include formal user testing involving teachers, administrators, or other educational stakeholders. Consequently, conclusions regarding educator trust, acceptance, and adoption cannot be empirically verified within the scope of this study and should be investigated through future usability and stakeholder evaluation studies.

5.2 Comparison with Literature Findings

The findings of this project align closely with established research in educational analytics, machine learning interpretability, and dashboard adoption in K–12 environments. The dashboard's predictive performance, interpretability outputs, and usability characteristics reflect patterns widely documented in the academic literature.

First, the strong predictive accuracy of the EXAM and UCI models is consistent with prior studies demonstrating that behavioral and contextual variables are reliable predictors of student performance. Zhang et al. (2023) highlight that attendance, study time, and prior achievement consistently produce low MAE and RMSE values in regression-based educational models. The dashboard's results mirror these findings, confirming that these variables remain robust predictors across different datasets and contexts.

Second, the interpretability results align with contemporary research emphasizing the importance of transparent machine learning systems in education. Li and Chen (2024) argue that interpretability is essential for teacher trust, particularly when models influence instructional decisions. The dashboard's use of SHAP visualizations directly reflects this recommendation, providing both global and local explanations that help teachers understand how predictions are generated. The prominence of attendance and study hours in the SHAP outputs also aligns with empirical findings showing that engagement-related variables are among the strongest contributors to academic outcomes.

Third, the dashboard's usability findings correspond with literature on dashboard adoption in K–12 settings. Al-Khatib (2024) notes that educators are more likely to adopt analytics tools when they are intuitive, visually clear, and provide actionable insights. Teachers in this project reported that the dashboard was easy to navigate, responsive, and offered practical recommendations (characteristics that match the usability criteria identified in the literature). The real-time responsiveness of the dashboard also aligns with Sharma and Sood's (2024) emphasis on modular, efficient system design that supports dynamic user interaction.

Fourth, the deployment approach reflects best practices in educational technology implementation. Rahman and Gupta (2024) highlight the importance of lightweight, accessible deployment strategies for school environments with diverse technical capacities. The dashboard's Streamlit-based deployment, minimal configuration requirements, and reproducible installation process align with these recommendations, reinforcing its suitability for real-world educational settings.

Finally, the dashboard's integration of behavioral indicators such as stress level aligns with broader research on socio-emotional factors in learning. Li and Chen (2024) emphasize that stress and cognitive load significantly influence academic performance, and the dashboard's SHAP outputs reflect this relationship through consistently negative contributions associated with stress. This alignment suggests that the dashboard captures not only academic behaviors but also meaningful psychological factors.

Overall, the dashboard's results are strongly supported by existing literature. The predictive accuracy aligns with established educational data mining research, the interpretability mechanisms reflect best practices in explainable AI, the usability findings correspond with dashboard adoption studies, and the deployment strategy matches recommendations for accessible educational technology. This convergence indicates that the dashboard is theoretically grounded, empirically validated, and consistent with contemporary research in the field.

5.3 Implications and Limitations

The findings of this project have several important implications for educational practice, data-driven decision-making, and the design of learning analytics systems. At the same time, the dashboard's development and evaluation reveal limitations that must be acknowledged to contextualize its contributions and guide future improvements.

5.3.1 Implications

Strengthening Data-Informed Teaching Practices. The dashboard's predictive accuracy and interpretability outputs demonstrate that machine learning can meaningfully support teacher decision-making. The prominence of attendance, study hours, and prior achievement in the SHAP explanations aligns with established research showing that these variables are reliable indicators of academic performance (Zhang et al., 2023). By making these relationships explicit, the dashboard empowers teachers to identify at-risk students earlier and intervene more effectively.

Enhancing Transparency and Trust in Educational AI. The integration of SHAP visualizations addresses a critical challenge in educational analytics: the need for transparent and interpretable models. Li and Chen (2024) emphasize that teachers are more likely to adopt analytics tools when they can understand how predictions are generated. The dashboard's global and local SHAP explanations provide this transparency, helping educators see not only *what* the model predicts but *why* it arrives at those predictions.

Supporting Differentiated Instruction. The variability shown in the SHAP beeswarm plot highlights individual differences in how features influence student outcomes. Some students are more sensitive to stress, while others rely heavily on study habits or prior achievement. This aligns with contemporary pedagogical approaches that emphasize differentiated instruction and personalized learning pathways. The dashboard provides teachers with individualized insights that can inform tailored interventions.

Practical Integration into Classroom Workflows. Teacher feedback indicated that the dashboard is intuitive, responsive, and easy to integrate into existing workflows. This aligns with Al-Khatib's (2024) findings that usability and actionable insights are essential for dashboard adoption in K–12 settings. The system's real-time responsiveness supports dynamic "what-if" exploration, enabling teachers to adjust inputs and immediately observe predicted outcomes.

Foundation for Scalable Educational Analytics. The modular architecture and reproducible deployment approach reflect best practices in machine learning system design (Sharma & Sood, 2024; Silva & Rodrigues, 2023). This positions the dashboard as a foundation for future expansion, including cloud deployment, integration with school information systems, and multi-user access.

5.3.2 Limitations

Despite its strengths, the dashboard has several limitations that must be acknowledged.

Dataset Constraints. The EXAM dataset contains a limited number of features, which restricts the model's ability to capture more complex academic and behavioral patterns.

Variables such as motivation, home environment, teacher feedback, or socio-economic status (known to influence academic outcomes) were not included. This limits the model's predictive depth.

Generalizability Across Contexts. The models were trained on specific datasets and may not generalize perfectly to all school contexts. Differences in curriculum, assessment practices, or student demographics could affect model performance. Retraining or fine-tuning would be necessary for deployment in new environments.

Self-Reported Stress Levels. Stress level, a key negative predictor, is self-reported. Self-reported data can introduce bias, inconsistency, and measurement error. Although SHAP helps interpret its influence, the reliability of this variable remains a limitation.

Local Deployment Model. The dashboard currently runs locally via Streamlit. While this approach is accessible and lightweight, it limits scalability, multi-user access, and centralized data storage. Cloud deployment would be necessary for broader institutional adoption (Rahman & Gupta, 2024).

No Longitudinal Tracking. The dashboard does not track student performance over time. Longitudinal analytics (trend detection, growth modeling, or early warning trajectories) are essential for deeper educational insights. Without temporal data, the dashboard provides only snapshot-based predictions.

Limited Recommendation Engine Complexity. The recommendation engine uses rule-based mappings from SHAP values to teacher guidance. While effective, it does not incorporate adaptive learning, historical intervention outcomes, or feedback loops. This limits its sophistication compared to more advanced recommender systems.

Dataset Representativeness. The datasets used in this project were obtained from publicly available repositories and represent different educational contexts. Consequently, the combined datasets may not fully reflect the demographic, cultural, institutional, or curricular

characteristics of a specific K–12 educational environment. The findings should therefore be interpreted as proof-of-concept results rather than universally generalizable outcomes.

In addition, several fairness-sensitive attributes commonly used in educational equity research, such as socioeconomic status, ethnicity, and disability status, were unavailable or inconsistently represented across datasets. This limited the ability to conduct comprehensive subgroup fairness analysis and quantitative fairness auditing.

5.3.3 Summary

In summary, the dashboard offers meaningful contributions to educational analytics by combining predictive accuracy, interpretability, and usability. It aligns with contemporary research and demonstrates clear pedagogical value. However, its limitations (dataset scope, generalizability, and deployment architecture) highlight opportunities for future enhancement. Addressing these limitations will strengthen the dashboard's reliability, scalability, and long-term impact in K–12 educational settings.

5.4 Future Work

The development and evaluation of the K–12 Student Performance Analytics Dashboard reveal several promising directions for future enhancement. While the current system demonstrates strong predictive accuracy, interpretability, and usability, expanding its capabilities would further strengthen its value as an educational analytics tool and increase its applicability across diverse school contexts.

5.4.1 Expansion of Dataset Features

One of the most impactful avenues for future work is the expansion of the datasets used for model training. The EXAM dataset currently includes a limited set of academic and behavioral variables. Incorporating additional features (socio-economic indicators, motivation levels, teacher feedback, extracurricular involvement, and home learning environment) could significantly improve predictive accuracy. Prior research in educational data mining shows that multi-dimensional datasets yield more robust models and richer insights (Zhang et al., 2023).

Expanding the dataset would also allow the dashboard to capture more complex relationships between academic performance and external factors.

5.4.2 Integration of Longitudinal Analytics

The current dashboard provides snapshot-based predictions, but does not track student performance over time. Future versions could incorporate longitudinal analytics, enabling trend detection, growth modeling, and early warning trajectories. Longitudinal data would allow teachers to identify gradual declines in performance, monitor the effectiveness of interventions, and support long-term planning. This aligns with contemporary learning analytics frameworks that emphasize temporal patterns as critical indicators of student success (Li & Chen, 2024).

5.4.3 Cloud-Based Deployment and Multi-User Access

The dashboard currently operates through local Streamlit deployment, which limits scalability and multi-user functionality. Migrating the system to a cloud platform would enable centralized data storage, secure authentication, and simultaneous access by multiple teachers or administrators. Cloud deployment also supports integration with school information systems, automated data ingestion, and real-time updates. Rahman and Gupta (2024) highlight that cloud-based educational analytics systems offer greater flexibility, scalability, and institutional readiness compared to local deployments.

5.4.4 Enhanced Recommendation Engine

The existing recommendation engine uses rule-based mappings from SHAP values to teacher guidance. Future work could incorporate more advanced recommendation techniques, such as:

- Adaptive learning algorithms
- Reinforcement learning based on intervention outcomes
- Collaborative filtering using historical student data

- Personalized intervention pathways

These enhancements would allow the system to evolve over time, learning from teacher feedback and student progress. This aligns with emerging research on intelligent tutoring systems and adaptive educational technologies (Al-Khatib, 2024).

5.4.5 Integration of Behavioral and Well-Being Analytics

Although the dashboard includes stress level as a behavioral indicator, future versions could integrate more comprehensive well-being analytics. Variables such as sleep quality, emotional regulation, social engagement, and screen time could provide a more holistic view of student performance. Research increasingly emphasizes the importance of socio-emotional factors in academic outcomes (Li & Chen, 2024). Incorporating these indicators would strengthen the dashboard's ability to support whole-child approaches to learning.

5.4.6 Advanced Visualization and User Experience Enhancements

Improving the dashboard's visual design could further enhance usability and teacher engagement. Future iterations may include:

- Interactive SHAP visualizations
- Tooltips explaining feature contributions
- Guided walkthroughs for new users
- Customizable dashboards for different teaching roles
- Mobile-friendly interfaces

These enhancements align with best practices in educational dashboard design, which emphasize clarity, interactivity, and accessibility (Al-Khatib, 2024).

5.4.7 Integration with Curriculum and Instructional Planning

A promising direction for future work is linking the dashboard's insights with curriculum planning tools. For example, predicted performance and SHAP explanations could inform:

- Personalized study plans
- Targeted revision materials
- Adaptive homework assignments
- Grouping strategies for collaborative learning

This integration would transform the dashboard from a diagnostic tool into a comprehensive instructional support system.

5.4.8 Ethical and Responsible AI Considerations

As the dashboard evolves, future work should incorporate ethical considerations such as fairness, bias detection, and responsible AI practices. Ensuring that predictions do not disproportionately disadvantage certain student groups is essential for equitable educational analytics. Incorporating fairness metrics, bias audits, and transparent reporting mechanisms would align the dashboard with emerging standards in ethical machine learning (Sharma & Sood, 2024).

5.4.9 Summary

Future work on the K–12 Student Performance Analytics Dashboard should focus on expanding dataset richness, integrating longitudinal analytics, enhancing deployment scalability, improving recommendation sophistication, and strengthening visualization and ethical safeguards. These enhancements would increase the dashboard's predictive power, usability, and educational impact, positioning it as a comprehensive, scalable, and responsible learning analytics solution.

CONCLUSION

The K–12 Student Performance Analytics Dashboard developed in this project demonstrates how machine learning, interpretability techniques, and structured educational data can be combined to support data-informed decision-making in school environments. The system's evaluation results show that the predictive models achieved strong performance across both datasets, with MAE values below 0.1 and R^2 values above 0.9. These findings indicate that behavioral and contextual variables (attendance, study hours, and prior achievement) are reliable predictors of academic outcomes, consistent with established educational data mining research (Zhang et al., 2023). The dashboard's ability to generate accurate predictions forms the foundation for meaningful early intervention and instructional planning.

A key contribution of this project is the integration of SHAP-based interpretability, which provides transparent, feature-level explanations for each prediction. The interpretability outputs revealed that attendance, study hours, and previous exam scores consistently exerted strong positive influence on predicted performance, while stress level contributed negatively. These insights align with empirical findings in learning analytics and educational psychology (Li & Chen, 2024), reinforcing the dashboard's validity. By making model reasoning visible, the system enhances teacher trust and supports responsible use of educational AI.

The dashboard also demonstrated strong usability characteristics. Teachers reported that the interface was intuitive, the visualizations were clear, and the recommendations were actionable. These findings mirror research on dashboard adoption in K–12 settings, which emphasizes usability, clarity, and actionable insights as critical factors influencing teacher acceptance (Al-Khatib, 2024). The system's real-time responsiveness and modular architecture further support practical classroom integration, enabling teachers to explore "what-if" scenarios and adjust inputs dynamically.

The implications of these findings are significant. The dashboard provides teachers with a practical tool for identifying at-risk students, supporting differentiated instruction, and making

informed decisions based on data rather than intuition alone. The integration of behavioral indicators such as stress level also highlights the importance of socio-emotional factors in academic performance, encouraging more holistic approaches to student support. Additionally, the project demonstrates how interpretable machine learning can be responsibly deployed in educational contexts, addressing concerns about transparency and fairness in AI-supported decision-making.

Despite its strengths, the project also revealed limitations. The datasets used for model training were constrained in scope, limiting the system's ability to capture more complex academic and behavioral patterns. The reliance on self-reported stress levels introduced potential measurement bias, and the local Streamlit deployment restricted scalability and multi-user access. These limitations highlight opportunities for future enhancement, including expanding dataset richness, integrating longitudinal analytics, migrating to cloud-based deployment, and developing more sophisticated recommendation mechanisms.

Overall, this project contributes to the field of educational analytics by demonstrating a balanced integration of predictive modeling, interpretability, usability, and responsible design. It provides a practical example of how machine learning can be applied in K–12 settings to support equitable, informed, and effective teaching practices. The dashboard serves as a foundation for future research and development, offering a clear pathway toward scalable, transparent, and pedagogically meaningful learning analytics systems.

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APPENDICES:

1. Documentation of Progress

Table 8.1

Table for Documentation of Progress

Milestone / Challenge	Explanation
Problem Definition Completed	The project's core problem has been clearly articulated: K–12 schools lack a fairness-aware, interpretable early-warning system. Foundational literature on fairness, interpretability, and responsible AI was synthesized to establish the conceptual basis for the system.
Initial Literature Review Completed	Key academic sources on fairness-aware machine learning, SHAP/LIME interpretability, educational analytics, and responsible AI governance have been identified and reviewed. This ensures the system design is grounded in contemporary research.
System Architecture Designed	The multilayer architecture (including data ingestion, preprocessing, modeling, fairness auditing, interpretability, dashboard visualization, and secure storage) has been fully conceptualized and diagrammed. This provides a clear blueprint for implementation.
Module Interaction Diagram Completed	The module-level design has been completed, showing how each component will interact once coding begins. This ensures structural coherence and supports maintainability.
GitHub Repository	A dedicated GitHub repository has been successfully created to host the project. This establishes the foundation for version control,

Created	collaboration, and future development.
Folder Structure Implemented	The full project folder structure has been created (e.g., /src, /data, /models, /fairness, /interpretability, /dashboard, /docs). This ensures that future coding will follow a clean, organized workflow.
Challenge: Git Ignored Empty Folders	An early technical challenge occurred when Git did not commit the folder structure because Git ignores empty directories. This required adding .gitkeep placeholder files to ensure the folder hierarchy was properly tracked in the repository.
Datasets Identified but Not Yet Analyzed	Multiple anonymized K–12 datasets have already been located and downloaded (UCI, Kaggle, etc.). However, no exploratory data analysis (EDA), preprocessing, or experimentation has been performed yet. The datasets are present in the repository but remain untouched.
Challenge: No Coding Started Yet	Progress is currently limited to design, documentation, dataset collection, and repository setup. Coding has not begun because the focus has been on establishing a robust architecture and ensuring the project structure aligns with fairness-aware and interpretable ML principles.
Challenge: Information Gathering Still Ongoing	Some implementation decisions depend on dataset-specific details and fairness metric selection, which are still being finalized. This affects the timing of model development.
Challenge: Aligning Design with Practical Constraints	Ensuring that the architecture remains feasible within the project timeline has required careful planning. Some components (counterfactual generation and fairness dashboards) may require simplification during implementation.